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Top-Two Concentration and Short-Price Protection Rules

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Developer Handover: Top-Two Concentration and Short-Price Protection Rules

Objective

Add a new market-structure rule family to the betting engine to identify races where the market is heavily concentrated in the top two runners, because these look materially more dangerous for standard lay logic.

This family should sit alongside the existing:

- core lay staking engine
- short-price control rules
- bookmaker vs exchange dislocation rules
- RPR overlay rules

This is not intended to replace the core engine. It is a protection and suppression layer.

The key idea is:

> If the top two runners in the betting are very strongly priced relative to the third runner, the race may effectively be a two-horse race. In these cases, normal favourite-lay logic should be reduced or blocked.

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1. Rule Family Name

Use:

TOP2_CONCENTRATION

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2. Priority

Recommended priority order in the engine:

1. `SHORT_PRICE_CONTROL`
2. `BMEX_ODDSON_DISLOCATION`
3. `TOP2_CONCENTRATION`
4. core lay engine
5. joint favourite / split logic
6. RPR overlay rules

Reason:

- short-price and bookmaker/exchange dislocation are more urgent
- top-two concentration is a market-structure protection layer
- RPR should not override a strong structural danger state

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3. Purpose

This rule family identifies races where:

- the top two runners in the current betting market dominate the win probability
- the third runner is materially weaker
- the race should therefore be treated as a concentrated top-two contest rather than a normal open market

When this happens, the engine should:

- suppress or block normal lay logic
- especially avoid laying both of the top two runners aggressively
- optionally reduce split exposure if joint-favourite logic is active

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4. Inputs Required

For the current market snapshot at each checkpoint:

Market ranking inputs

- `runner_1_id`
- `runner_1_exchange_back_odds`
- `runner_2_id`
- `runner_2_exchange_back_odds`
- `runner_3_id`
- `runner_3_exchange_back_odds`

These are the current 1st, 2nd, and 3rd in betting order by exchange back odds.

Derived probabilities

Use decimal odds converted to implied probability:

p1 = 1 / odds_1
p2 = 1 / odds_2
p3 = 1 / odds_3

Derived concentration metrics

Store:

top2_combined = p1 + p2
third_vs_second_ratio = p3 / p2
second_vs_first_ratio = p2 / p1

Optional additional metric:

top2_vs_top3_share = (p1 + p2) / (p1 + p2 + p3)

This can be useful later, but is not required for day one.

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5. Checkpoints

Run this family at:

- `T-30`
- `T-15`
- `T-5`

- `T-1`

Most useful checkpoints are:

- `T-15`
- `T-5`
- `T-1`

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6. Core Concepts

A. Top-two combined implied probability

This measures how much of the win market is concentrated in the top two runners.

Example:

- Runner 1 = 11/10 = 2.10 decimal → `0.4762`
- Runner 2 = 6/5 = 2.20 decimal → `0.4545`

Then:

top2_combined = 0.4762 + 0.4545 = 0.9307

That is extremely concentrated.

B. Third-vs-second ratio

This measures how weak the third runner is relative to the second.

Example:

- Third = 10/1 = 11.00 decimal → `0.0909`
- Second = 2.20 decimal → `0.4545`

Then:

third_vs_second_ratio = 0.0909 / 0.4545 = 0.20

That is a very large gap and suggests a strongly compressed top two.

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7. Day-One Thresholds

These are starting defaults for testing, not fixed truths.

Top-two combined bands

- mild concentration: `top2_combined >= 0.60`
- medium concentration: `top2_combined >= 0.65`
- strong concentration: `top2_combined >= 0.70`
- extreme concentration: `top2_combined >= 0.80`

Third-vs-second ratio bands

- mild third gap: `third_vs_second_ratio <= 0.60`
- medium third gap: `<= 0.50`
- strong third gap: `<= 0.40`
- extreme third gap: `<= 0.30`

Second-vs-first ratio

This is optional but useful for seeing if the top two are close enough to behave like co-leaders.

Suggested bands:

- close top two: `second_vs_first_ratio >= 0.85`
- very close top two: `>= 0.92`

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8. Rule Outputs

This family should return a state object like:

```
{
  "rule_family": "TOP2_CONCENTRATION",
  "is_active": true,
  "severity": "strong",
  "top2_combined": 0.93,
  "third_vs_second_ratio": 0.20,
  "second_vs_first_ratio": 0.95,
  "lay_action": "SUPPRESS",
  "lay_multiplier": 0.25,
  "allocation_cap_active": true,
  "reason_codes": [
    "TOP2_COMBINED_STRONG",
    "THIRD_GAP_EXTREME",
    "TOP2_CLOSE_TOGETHER"
  ]
}
```

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9. Rule Definitions

TOP2_01_SCOPE

Purpose: identify races with at least 3 valid runners in market order.

Run at: T-30, T-15, T-5, T-1

Preconditions:

- at least 3 priced runners available
- valid exchange back odds for top 3 runners

Effect:

- enables rule family calculations

TOP2_02_TOP2_COMBINED_CONCENTRATION

Purpose: classify how much of the market is concentrated in the top two.

Run at: T-30, T-15, T-5, T-1

Calculations:

```
p1 = 1 / odds_1
p2 = 1 / odds_2
top2_combined = p1 + p2
```

Bands:

- mild: `>= 0.60`
- medium: `>= 0.65`
- strong: `>= 0.70`
- extreme: `>= 0.80`

Output:

- `top2_combined`
- severity band

TOP2_03_THIRD_GAP_CONFIRMATION

Purpose: confirm whether the third horse is materially weaker than the second.

Run at: T-30, T-15, T-5, T-1

Calculation:

third_vs_second_ratio = p3 / p2

Bands:

- mild gap: `<= 0.60`
- medium gap: `<= 0.50`
- strong gap: `<= 0.40`
- extreme gap: `<= 0.30`

Output:

- `third_vs_second_ratio`
- gap severity

TOP2_04_TOP2_CLOSE_TOGETHER

Purpose: identify when the first and second in betting are both strong and close enough to each other that the race behaves more like a two-horse market.

Run at: T-30, T-15, T-5, T-1

Calculation:

second_vs_first_ratio = p2 / p1

Bands:

- close top two: `>= 0.85`
- very close top two: `>= 0.92`

Output:

- `second_vs_first_ratio`
- closeness band

TOP2_05_MILD_WARNING_STATE

Purpose: warning state only, no hard action yet.

Run at: T-15, T-5, T-1

Trigger logic:

top2_combined >= 0.60
AND third_vs_second_ratio <= 0.60

Effect:

lay_action = "WATCH"
lay_multiplier = 1.00

No stake reduction yet. Logging only.

TOP2_06_MEDIUM_SUPPRESSOR

Purpose: reduce lay size in moderately concentrated top-two markets.

Run at: T-15, T-5, T-1

Trigger logic:

top2_combined >= 0.65
AND third_vs_second_ratio <= 0.50

Effect:

lay_action = "SUPPRESS"
lay_multiplier = 0.60

Interpretation:

- keep 60% of normal lay size

TOP2_07_STRONG_SUPPRESSOR

Purpose: aggressively reduce lay size in strong top-two markets.

Run at: T-5, T-1

Trigger logic:

top2_combined >= 0.70
AND third_vs_second_ratio <= 0.40

Effect:

lay_action = "SUPPRESS"
lay_multiplier = 0.25

Interpretation:

- keep only 25% of normal lay size

TOP2_08_TWO_HORSE_RACE_BLOCK

Purpose: hard block or near-block for extreme two-horse race structures.

Run at: T-5, T-1

Trigger logic:

top2_combined >= 0.80
AND third_vs_second_ratio <= 0.30
AND second_vs_first_ratio >= 0.85

Effect:

lay_action = "BLOCK"
lay_multiplier = 0.00

Interpretation:

- treat as a two-horse race
- no standard lay allowed

TOP2_09_SPLIT_EXPOSURE_CAP

Purpose: if the engine is already in a split / joint-favourite state, cap how much exposure can be distributed across the top two in a concentrated market.

Run at: T-15, T-5, T-1

Preconditions:

- joint favourite or split state active

Trigger logic:

top2_combined >= 0.65
AND third_vs_second_ratio <= 0.50

Effect:

- cap total split exposure across top two
- recommended:

max_total_points_across_top2 = 0.50 × normal_split_exposure

This should stop the system from laying both top runners too aggressively in a market that is clearly concentrated at the top.

TOP2_10_PERSISTENCE_CONFIRMATION

Purpose: confirm the concentration state is not fleeting.

Run at: T-15, T-5, T-1

Trigger logic:

- same or very similar top two remain top two
- `TOP2_06`, `TOP2_07`, or `TOP2_08` active at current and previous checkpoint

Effect:

- increase confidence in suppressor/block decision

This can be required for hard blocking if preferred.

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10. Resolution Logic

The family should resolve to one of these states:

STATE A: NONE

No meaningful concentration.

Effect:

- no change to core lay logic

STATE B: WATCH

Mild top-two concentration.

Effect:

- log only
- no change

STATE C: SUPPRESS_MEDIUM

Moderate top-two concentration.

Effect:

- `lay_multiplier = 0.60`

STATE D: SUPPRESS_STRONG

Strong top-two concentration.

Effect:

- `lay_multiplier = 0.25`

STATE E: BLOCK

Extreme two-horse-race state.

Effect:

- `lay_multiplier = 0.00`

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11. Integration with Core Engine

Step 1

Run `SHORT_PRICE_CONTROL`.

If blocked there, stop. No lay.

Step 2

Run `BMEX_ODDSON_DISLOCATION`.

If blocked there, stop. No lay.

Step 3

Run `TOP2_CONCENTRATION`.

Step 4

If `TOP2_CONCENTRATION` returns:

- `WATCH` → no change
- `SUPPRESS_MEDIUM` → multiply base lay points by `0.60`

- `SUPPRESS_STRONG` → multiply base lay points by `0.25`
- `BLOCK` → set lay points to zero

Step 5

Only after that, apply RPR overlay to any remaining lay exposure.

Important:

- RPR may scale remaining exposure slightly
- RPR must not override a `BLOCK` from `TOP2_CONCENTRATION`

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12. Examples

Example 1: screenshot-style market

Approx:

- odds_1 = 2.10
- odds_2 = 2.20
- odds_3 = 11.00

Then:

p1 = 0.4762
p2 = 0.4545
p3 = 0.0909

top2_combined = 0.9307
third_vs_second_ratio = 0.20
second_vs_first_ratio = 0.95

This should trigger:

- `TOP2_02`: extreme
- `TOP2_03`: extreme
- `TOP2_04`: very close top two
- `TOP2_08`: BLOCK

Example 2: more normal open race

- odds_1 = 3.20
- odds_2 = 4.20
- odds_3 = 5.50

Then:

p1 = 0.3125
p2 = 0.2381
p3 = 0.1818

top2_combined = 0.5506
third_vs_second_ratio = 0.76
second_vs_first_ratio = 0.76

This should produce:

- no top-two concentration action

Example 3: moderate top concentration

- odds_1 = 2.70
- odds_2 = 3.10
- odds_3 = 8.00

Then:

p1 = 0.3704
p2 = 0.3226
p3 = 0.1250

top2_combined = 0.6930
third_vs_second_ratio = 0.39
second_vs_first_ratio = 0.87

This should likely trigger:

- `SUPPRESS_STRONG`

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13. Logging Requirements

For every market at every checkpoint, store:

- `race_id`
- `timestamp`
- `odds_1`, `odds_2`, `odds_3`
- `p1`, `p2`, `p3`
- `top2_combined`
- `third_vs_second_ratio`
- `second_vs_first_ratio`
- active `TOP2_CONCENTRATION` state
- final lay multiplier after this family
- race winner rank in betting at checkpoint
- P&L under:
  - core lay only
  - core lay + top2 concentration suppression
  - core lay + top2 concentration block

This is important because the exact thresholds will probably need calibration from your own data.

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14. Recommended Day-One Launch

I would launch only these rules first:

- `TOP2_01_SCOPE`
- `TOP2_02_TOP2_COMBINED_CONCENTRATION`
- `TOP2_03_THIRD_GAP_CONFIRMATION`
- `TOP2_04_TOP2_CLOSE_TOGETHER`
- `TOP2_06_MEDIUM_SUPPRESSOR`
- `TOP2_07_STRONG_SUPPRESSOR`
- `TOP2_08_TWO_HORSE_RACE_BLOCK`

That gives a clean first version without overcomplicating persistence and split caps.

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15. Short Implementation Summary

For the developer:

1. Rank runners by current exchange back odds.
2. Take top 3.
3. Convert top 3 odds into implied probabilities.
4. Calculate:
   - top2_combined = p1 + p2
   - third_vs_second_ratio = p3 / p2
   - second_vs_first_ratio = p2 / p1
5. Apply thresholds:
   - if top2_combined >= 0.80 and third_vs_second_ratio <= 0.30 and second_vs_first_ratio >= 0.85:
     BLOCK
   - else if top2_combined >= 0.70 and third_vs_second_ratio <= 0.40:
     SUPPRESS_STRONG (0.25)
   - else if top2_combined >= 0.65 and third_vs_second_ratio <= 0.50:
     SUPPRESS_MEDIUM (0.60)
   - else:
     no action
6. Apply this result before RPR overlays.

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